

Description_and_Outline

October 31, 2025

1 Project Description

Link to App <https://datascience-7zcp6hewugkdo4dda6hh.streamlit.app/>

I partnered with coach John Andrzejek on an analytics project involving player valuation in college basketball. Coach Andrzejek is the current head coach at Campbell University, and was previously an assistant at Florida, Washington State, and the University of San Francisco (which is how I had a connection to him through my dad, who teaches at USF).

The goal of the project was to create a way to properly value players on the team from two perspectives.

First, in the new era of NIL collectives, we want to know how much a player's performance contributes to his team's success for proper compensation.

Secondly, the team wants to understand the value added by players in an attempt to develop and improve the team.

Example: Campbell is currently ranked 203rd in NCAA DI by Ken Pomeroy: <https://kenpom.com/>. Their goal is to move up to 140th this season. The app I built lets a coach analyze the player value profiles of a typical 203rd ranked team and compare them to a 140th ranked team. This can help quantify and clarify issues around recruitment, use of the transfer portal, and player development.

The app also allows for exploration of questions like:

- How much would the team improve if our number one (or two, or three, etc.) player had a particular value?
- How is the distribution of player value over the top 10 players different between a 203 team and a 140 team?

and other related questions.

Question: How is value measured?

The starting point is a traditional basketball metric called Win Shares (WS). The details of the WS calculation are given here <https://www.sports-reference.com/cbb/about/ws.html>. The choice of WS as a value metric was made by Coach Andrzejek since this is a statistic that is familiar and intuitive to most coaches working in college basketball.

The basic idea of WS is that the sum of a team's players' WS should be about equal to the number of games the team won. WS is calculated in two parts: Offensive Win Shares (OWS) and Defensive Win Shares (DWS), with $WS = OWS + DWS$. The key ingredient in a player's OWS is a metric

called Offensive Rating (ORTg). A player's ORtg is an estimate of the number of points the player created per 100 possessions.

Another relevant metric is USG%, which measures the percentage of a team's possessions *used* by a particular player (by shots, free throws, or turnovers). The relationship between USG% and ORtg% has been widely discussed in sports analytics, usually in the context of the NBA.

One of our claims is that in college basketball, high usage players who maintain efficiency are *especially* valuable. A high usage percentage (with good efficiency) reflects, among other things, a player's ability to create offense which is a high-value skill.

Much of the early work in this project is devoted to analyzing the relationship between usage (USG%) and efficiency (ORTg), and making adjustments to properly reflect player value in college (without changing the underlying interpretation of Win Shares). The result is a new metric we call virtual Win Shares (vWS). We then use vWS to build the app discussed above.

Spoiler: Our claim is supported by the fact that our adjusted statistic called virtual Win Shares (vWS) correlates significantly better with team success than traditional Win Shares.

2 Project Outline

2.1 I. Scrape Data

1. Scrape all relevant player data from Sports Reference: <https://www.sports-reference.com/cbb/> and merge into a comprehensive dataset of all players from 2011-2025. (Separate file with code)

2.2 II. Develop and validate the vWS metric.

1. We introduce vWS as a modification of the traditional Win Shares (WS) metric that better reflects value in college basketball. The development process is detailed with justifications, visualizations, and analysis in Notebook2.
2. Calculate vWS for each player from 2011 to present.
3. Validate vWS by showing that vWS (aggregated over all players on a team) correlates significantly more with team success than traditional (aggregate) WS. Team success is measured by (team) Adjusted Efficiency Margin as reported by <https://kenpom.com/>.

2.3 III. Prepare Team Level Data

1. Convert the player-level vWS data to a wide format where each row represents a team in a season and columns give vWS values for the players on the team in order of their vWS ranking (1st, 2nd, 3rd, etc.).

2.4 IV. Build App

1. Build models and visualizations on the dataset to facilitate coach's use and analysis. Wrap these models into an app that allows for easy interaction and reporting.
2. The models are simple ols regression models to predict the vWS of the i -th ($1 \leq i \leq 12$) ranked player on a team with user inputted KenPom AdjEM ranking.

Notes:

- II, and III are each presented in a separate jupyter notebooks. PDFs of code/output are included.
- II contains the mathematical and statistical details in the vWS development process and represents the largest time investment on the project.
- Code for the web scraper and data/merge in I is on github.
- Code for IV is on github along with the required datasets. While I used minimal AI assistance in I-III, I relied heavily on AI for coding assistance in building the web app with Streamlit. (That is probably pretty obvious.) The focus of this part of the project was carefully prompting so that the resulting visualizations were clear and useful to the coaching staff.

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